

# Engineering, Remote Sensing & Geospatial Analysis

Forest Engineer with a Ph.D. in Remote Sensing from INPE, bringing over 15 years of expertise in geospatial analysis, geoprocessing, and the development of web-based solutions for environmental management and infrastructure.



## Summary

Vinicius do Prado Capanema holds a Bachelor's degree in Forest Engineering, a specialization in Rural Property Georeferencing accredited by INCRA, a Master's and a Ph.D. in Remote Sensing, and an MBA in Project Management. With a multidisciplinary background, he operates in an integrated manner across research, development, and innovation in the road infrastructure sector. He has extensive experience in conceiving, building, and deploying technology-driven solutions that leverage geoprocessing, geostatistics, and remote sensing for data analysis and manipulation, with emphasis on cloud computing, geographic databases, and spatial data infrastructure (SDI).

**Consolidated experience in applying Artificial Intelligence to geospatial data management and analysis**, with focus on process automation, operational efficiency improvement, and strategic decision support. Works with generative AI as a catalyst for technical productivity, integrating advanced models into geoprocessing routines, remote sensing, data governance, and spatial data engineering. Strong capability in structuring intelligent workflows, optimizing pipelines, qualifying information, standardizing procedures, and enhancing analytical and documentary production. Leverages AI as an executive tool to elevate delivery quality, reduce response time, support complex decision-making, and drive innovation in multidisciplinary, data-driven environments.

By applying advanced geoprocessing techniques to remote sensing-derived data, he conducts studies focused on monitoring and mapping immediate and adjacent factors that may impact road infrastructure development and management, providing decision support for strategic planning.

Beyond technical research, he contributes to multidisciplinary projects involving engineering, the environment, and technology, fostering the integration of interdisciplinary teams to ensure effective project delivery. His expertise in forest engineering and remote sensing enables him to perform environmental impact analyses, develop computational models, and create decision-support tools, with integration into cloud architectures and corporate spatial data ecosystems, contributing to project sustainability.

Alongside operational activities, he is dedicated to knowledge dissemination through the preparation and publication of scientific studies, technical reports, and active participation in conferences, seminars, and training sessions, strengthening best practices and encouraging the continuous improvement of applied methodologies.

## Core Competencies

### Artificial Intelligence & Geospatial Governance

**AI Applied to Geospatial Data:** Consolidated experience in applying Artificial Intelligence for automating and optimizing processes in geoprocessing and remote sensing. Integration of generative AI models into geospatial analysis routines, enhancing technical productivity and delivery quality.

**Spatial Data Governance:** Structuring intelligent data workflows, optimizing ETL/ELT pipelines, qualifying and standardizing geospatial information. Development of procedures to ensure consistency, traceability, and reliability of data in corporate spatial data ecosystems.

**AI as Executive Tool:** Application of AI to strategic decision support, reducing response time, analyzing complex scenarios, and driving innovation in multidisciplinary environments. Leveraging advanced models to support complex decision-making and drive digital transformation in data-driven organizations.

### Research & Development

**Researcher Profile:** Ph.D. in Remote Sensing from INPE with a solid scientific track record (peer-reviewed journal articles, conference presentations, peer review). Expertise in developing innovative methodologies for geospatial analysis and environmental monitoring.

**R&D Leadership:** Proven ability to lead and coordinate multidisciplinary research groups, from project conception through publication. Experience in fundraising, research timeline management, and building specialized technical teams.

**Technological Innovation:** Development of applied tools and methodologies (FAD-AHP, SIGMA-PLI, WebGIS, indicator calculators), AI integration in research workflows, and automation of analytical processes.

## Project & Team Management

**Project Coordination:** Planning and execution of complex environmental studies involving multiple stakeholders (DER/SP, SEMIL, environmental agencies). Scope, budget, and deliverable management with a focus on measurable outcomes.

**Team Leadership:** Proven experience in coordinating multidisciplinary teams (engineers, geographers, data analysts, developers), fostering a collaborative environment and efficient task distribution.

**Institutional Liaison:** Articulation with regulatory bodies (INCRA, IBAMA, SEMA-MT, CETESB), presenting results to decision-makers, and preparing technical reports and opinions for environmental licensing.

## Environmental Governance & Sustainability

**Environmental Policies:** Structuring environmental governance frameworks, strategic alignment with decarbonization targets (SP Carbono Zero), development of sustainability indicators, and regulatory compliance monitoring.

**Risk Management:** Climate risk analysis and mitigation, natural disaster prevention, socio-environmental vulnerability studies, and planning of preventive and corrective actions.

**Licensing & Compliance:** Oversight of environmental licensing processes (LP, LI, LO), preparation of EIA/RIMA, PRAD, PCA, management of environmental requirements, and liaison with regulatory authorities.

## Geoprocessing & Remote Sensing

**Advanced Spatial Analysis:** Satellite image processing (Sentinel, Landsat, MODIS, CBERS) in Google Earth Engine, QGIS, ArcGIS Pro; supervised/unsupervised classification, object-oriented segmentation, and time series analysis.

**Spectral Indices:** Calculation and interpretation of NDVI, EVI, NDWI, and NBR for vegetation monitoring, water resources assessment, and fire scar detection; validation with field data.

**Practical Applications:** Land use and land cover mapping, multitemporal change detection, forest degradation monitoring, urban sprawl analysis, and infrastructure asset inventory.

## Geospatial Data Engineering

**Data Modeling:** Design of optimized PostGIS schemas, structure normalization, definition of constraints and geometric validations, spatial indexes (GIST, SP-GIST), and large table partitioning.

**ETL/ELT Pipelines:** Development of automated extraction, transformation, and loading workflows for geospatial data (IBGE/SIDRA, SECEX, OpenStreetMap, satellite imagery), orchestration with Airflow/Python, logging and monitoring.

**Source Integration:** Consolidation of heterogeneous data (shapefiles, GeoJSON, raster, CSVs, REST APIs), CRS transformations, geometry cleaning and validation, and enrichment with auxiliary datasets.

## WebGIS & Application Development

**Geospatial Frontend:** Development of interactive interfaces with Leaflet.js (layer control, dynamic filters, custom popups, clustering, heatmaps, drawing tools), integration with REST APIs.

**Backend & APIs:** Creation of geospatial APIs with FastAPI (endpoints for spatial queries, buffers, intersections, geocoding), JWT authentication, Swagger/OpenAPI documentation, and performance optimization.

**OGC Services:** Publishing WMS/WFS/WMTS via GeoServer and QGIS Server, SLD styling configuration, caching with GeoWebCache, and integration with PostGIS for dynamic data.

## Multi-Criteria Analysis & Decision Support

**AHP Method (Saaty):** Expertise in Analytic Hierarchy Process for criteria weighting and decision support (applied in DER/SP road jurisdictionalization projects, favorability indicators).

**Indicators & Metrics:** Development of indicator adherence evaluation systems, multi-criteria classification (Direct, Strong Proxy, Moderate Proxy, Uncertain), and methodological validation with traceability.

**Support Tools:** Creation of interactive calculators and dashboards for scenario analysis, impact simulations, and result visualization for non-technical stakeholders.

# Languages

**English** — Advanced (reading, writing, conversation)

**Spanish** — Intermediate

## Courses & Certifications

### → Georeferencing & Surveying

- 3rd Edition of the Technical Standard for Rural Property Georeferencing (NTGIR) and SIGEF — ANOREG-MT
- Official TopoEVN 6 Surveying Certification — Metrica
- AutoCAD | TOPOEVN — Professional training course
- Technical responsibility in land tenure processes — INTERMAT-MT

### → Geoprocessing & GIS

- Orbital data processing in GIS platforms (ENVI, SPRING, QGIS, ArcGIS, TERRAVIEW, GEODA, TerraAmazon)
- Geoprocessing | Spatial Analysis | R Programming
- Land use and land cover change modeling with Dinamica EGO — UFMG (2018)

### → Programming & Data Science

- Python for Remote Sensing Research — INPE, São José dos Campos
- Microsoft Office (Word, Excel, PowerPoint, Internet) — Advanced course

### → Environment & Licensing

- Rural Environmental Registry (CAR) preparation — SEMA-MT
- Technical responsibility in environmental processes — SEMA-MT
- Phytogeography — UNEMAT (2008)

### → Management & Business

- Business strategy | Customer service
- Project planning | Project management

## Projects

### → **Satellite-Based Environmental Monitoring (INPE)**

2017–2018

- Study of Pattern and Process Trajectories in Characterizing Deforestation Dynamics in the Amazon.
- Project aimed at supporting the expansion and improvement of satellite-based environmental monitoring and the development of land use and land cover studies in the Amazon region.
- Contribution to the implementation of the country's REDD (Reducing Emissions from Deforestation and Forest Degradation) strategy, enabling greater control of deforestation and forest degradation through the strengthening of INPE's monitoring systems.

### → **e-Sensing: Big Earth Observation Data Analytics**

Doctoral Research (2018–2022)

- Project with the primary objective of conceiving, building, and deploying a new type of knowledge platform for organizing, accessing, processing, and analyzing large-scale Earth observation data.
- The knowledge platform enabled scientists to produce information in innovative ways, based on open-source software, allowing data sharing and result reproducibility.
- Focus on improving the extraction of land use and land cover change information using advanced big data analytics and machine learning techniques.

## Technologies & Tools

### → Geoprocessing & Remote Sensing

**Desktop GIS:** QGIS (graphical modeler, Python plugins, advanced spatial analysis), ArcGIS/ArcGIS Pro (geoprocessing, network analysis, 3D), TerraView, TerraAmazon (forest monitoring), GEODA (exploratory spatial analysis), AutoCAD (CAD-GIS projects), TopoEVN (surveying and georeferencing).

**Cloud & Big Data GEO:** Google Earth Engine (massive satellite image processing, time series, supervised/unsupervised classification, spectral indices NDVI/EVI/NDWI).

**Libraries & CLI:** GDAL/OGR (raster/vector conversion and processing via command line and Python), PROJ (coordinate transformations), Rasterio, Fiona, Shapely, GeoPandas.

### → Programming & Data Analysis

**Python:** Pandas/GeoPandas (tabular and spatial data manipulation), NumPy/SciPy (scientific computing), Matplotlib/Seaborn (visualization), Scikit-learn (machine learning for image classification), Rasterio/GDAL (raster processing), Shapely/Fiona (vector geometries), PyProj (projections), Requests/BeautifulSoup (web scraping), SQLAlchemy (database ORM).

**R:** Spatial statistical analysis, satellite image processing (raster, terra, rgdal), visualization (ggplot2), predictive modeling, time series analysis.

**JavaScript:** Intermediate proficiency in web development, DOM manipulation, REST API integration, visualization libraries (Chart.js, Leaflet.js).

### → Spatial Databases

**PostgreSQL/PostGIS:** Spatial data modeling, ST\_\* functions (ST\_Intersects, ST\_Buffer, ST\_Union, ST\_Distance, ST\_Within), spatial indexes (GIST, SP-GIST, BRIN), topological analysis, reverse geocoding, spatial query optimization, triggers, and stored procedures.

**Best Practices:** Schema normalization, geometric constraints and validations, materialized views for complex queries, large table partitioning, backup and restore, replication and high availability.

### → WebGIS & Service Publishing

**Frontend:** Leaflet.js (interactive maps, layer control, dynamic filters, custom popups, clustering plugins, heatmaps, drawing tools), OpenLayers (alternative for complex projects).

**Map Servers:** GeoServer (WMS/WFS/WMTS publishing, SLD styling, GeoWebCache caching, PostGIS integration), QGIS Server (lightweight OGC services), MapServer.

**Tiles & Performance:** Vector tile generation (MVT), raster tile caching, style optimization, data compression, CDN for global delivery.

## → APIs & Geospatial Backend

**FastAPI:** Development of geospatial REST APIs (endpoints for spatial queries, geometry filters, dynamic buffers), JWT authentication, rate limiting, structured logging, Pydantic validation, automatic documentation (Swagger/OpenAPI).

**Optimization:** Optimized SQL queries with EXPLAIN ANALYZE, materialized views, Redis caching, result pagination, GeoJSON response compression, configured CORS.

**Integration:** Consumption of external APIs (IBGE, INPE, OpenStreetMap, weather services), webhooks, asynchronous processing with Celery/RQ.

## → DevOps & Infrastructure

**Containers:** Docker (image creation, optimized Dockerfile, multi-stage builds), Docker Compose (local orchestration of full stacks: PostGIS + GeoServer + API + Frontend).

**Web Servers:** Nginx (reverse proxy, load balancing, gzip/brotli compression, HTTP caching, SSL/TLS configuration), Apache (alternative for legacy projects).

**Deploy & CI/CD:** Git/GitHub (version control, workflows), knowledge of GitHub Actions, deployment on Linux servers (Ubuntu/Debian), basic monitoring with logs.

## → AI & Applied Machine Learning

**AI Tools:** Proficiency in modern artificial intelligence tools (ChatGPT, Claude, GitHub Copilot) applied to code generation, data analysis, process automation, technical documentation, and complex problem-solving.

**Geospatial ML:** Supervised satellite image classification (Random Forest, SVM), change detection, object segmentation, spatial predictive analysis, model validation (confusion matrix, accuracy, Kappa).

**Natural Language Processing:** Prompt engineering for task optimization, LLM integration in development workflows.

## → Data Governance & Quality

**Metadata & Cataloging:** Dataset documentation following ISO 19115/19139 standards, GeoNetwork for cataloging, data lineage tracking (provenance).

**Version Control:** Git for code and scripts, shapefile/GeoJSON version control, database change history (audit triggers).

**Quality:** Topological geometry validation, overlap/gap detection, completeness analysis, positional accuracy, reproducible documentation with Jupyter Notebooks/R Markdown.

## → Office & Productivity

**Microsoft Office:** Word (complex technical report preparation, ABNT formatting, collaborative reviews), PowerPoint (executive presentations with infographics, maps, and dashboards), Excel (data analysis with pivot tables, advanced formulas, VBA macros, Power Query integration), Outlook (email, calendar, and task management).

**Other Tools:** Google Workspace (Docs, Sheets, Slides for real-time collaboration), Markdown (technical documentation), LaTeX (scientific documents), Notion/Obsidian (knowledge management).

## Education

### Ph.D.

Remote Sensing · INPE · 2022

### MBA

Project Management · Unopar · 2020

### Master's Degree

Remote Sensing · INPE · 2017

### Specialization

Georeferencing · F. Vale do Juruena · 2010

### Bachelor's Degree

Forest Engineering · UNEMAT · 2009

## Professional Experience

### SEMIL/SP — Data Engineer

July 2025 – Present

- Develops web-based solutions applying geoprocessing and geostatistical data analysis for the Integrated Logistics Plan (PLI).
- Supports the construction and development of PLI products focused on infrastructure and logistics management for the State of São Paulo.
- Applies advanced spatial data processing techniques to support strategic decision-making.
- Integrates data from multiple sources for the creation of dashboards and territorial analysis tools.

### DER/SP — Interim Environmental Coordinator

Sep 2024 – Jan 2025

- Led the coordination of environmental governance and sustainability initiatives, structuring guidelines and aligning policies with decarbonization targets (SP Carbono Zero).
- Coordinated strategic environmental studies for infrastructure management, decarbonization, climate risk assessment, and disaster prevention, applying advanced geoprocessing and remote sensing techniques.
- Defined data standards and workflows for environmental licensing and preservation programs; integrated dashboards and thematic maps for decision-making.
- Oversaw environmental licensing processes ensuring regulatory compliance; planned and executed preventive and corrective actions for environmental impact mitigation.
- Actively participated in formulating environmental policies, best practices, and regulations applied to infrastructure and territorial management.

## **DER/SP — Forest Engineer & GIS Specialist**

Aug 2023 – Sep 2024

- Performed maintenance and management of the state road network through satellite imagery, reconciling topology and attributes with administrative sources.
- Managed road infrastructure assets for socio-environmental adequacy and compliance; implemented compliance routines and partner integrations.
- Coordinated research and studies aimed at road network maintenance and improvement; developed spatial analyses and technical reports.
- Documented procedures and metadata; supported R&D focused on automation and geospatial data quality improvement.

## **Stantec — Technical Consultant (Remote - Belgium)**

Jan 2024 – Mar 2024

- Engaged in an international technical consulting project providing remote services for operations in Belgium.
- Performed collection, processing, and analysis of geospatial data involving remote sensing.
- Applied advanced geoprocessing techniques for spatial data analysis within a European context.

## **VEGA Monitoring — Geoprocessing Analyst**

Mar 2022 – Aug 2023

- Performed grain yield modeling at the national level; conducted meta-analyses of sector-specific data for agribusiness decision-making.
- Developed and managed geospatial databases of cultivated areas for client decision support; generated risk area data for crop planning and commodity trading.
- Developed ESG and sustainability projects using remote sensing; validated results and produced technical reports.
- Specialist in the R&D coordination unit, pursuing technologies and methodologies to optimize remote sensing, geoprocessing, and modeling processes.

## **GeoBrasilis — GIS Specialist Consultant**

Sep 2021 – Mar 2022

- Monitored urban sprawl expansion around the Rodoanel Mário Covas through multitemporal satellite image classification.
- Produced technical reports with emphasis on urban monitoring; performed classification and accuracy assessment.

## **FUNCATE — Geoprocessing Specialist**

May 2017 – Mar 2018

- Worked on a satellite-based environmental monitoring project for the Amazon biome; produced and analyzed remote sensing data for forest degradation studies.
- Generated thematic land use and land cover maps using geoprocessing, spatial analysis, and digital image processing techniques.
- Organized, planned, and managed field campaigns and teams for qualitative land use and land cover data collection in the Amazon.

## **Mognos Forest & Civil Engineering — Forest Engineer**

2010 – 2015

- Planned territorial and environmental management for rural enterprises; developed and executed Degraded Area Recovery Plans (PRAD) for permanent preservation areas and legal reserves.
- Coordinated multidisciplinary teams for environmental monitoring and licensing; developed projects to optimize client service.
- Prepared and managed administrative processes with INCRA (CCIR, ITR, georeferencing), IBAMA, SEMA-MT, and INTERMAT-MT.
- Managed forest inventory in native forests, environmental licensing and regularization of industries and rural properties; prepared the Rural Environmental Registry (CAR).
- Performed geoprocessing and georeferencing in compliance with INCRA technical standards; managed PMFS and PEF; prepared technical reports and EIA/RIMA.
- Developed maps using GIS (QGIS, ArcGIS, ENVI, TerraView) for quantification and study of the spatiotemporal dynamics of deforestation and forest fires.

## Publications

### → **Characterization of municipalities in the timber production chain**

XXV IUFRO World Congress · 2019

CAPANEMA, V. P.; ESCADA, M. I. S. Characterization of municipalities in the timber production chain in the north of Mato Grosso, Brazil. In: IUFRO World Congress, Curitiba, PR. 2019.

### → **Landscape patterns associated with extractivism in the Amazon**

XXV IUFRO World Congress · 2019

SOUZA, A. R.; ESCADA, M. I. S.; SANTOS, G. V. S.; MONTEIRO, A. M. V.; CAPANEMA, V. P. Landscape patterns associated with extractivism in the Amazon: characterization and mapping of potential açaí occurrence areas in northeastern Pará. In: IUFRO World Congress, Curitiba, PR. 2019.

### → **Comparison between TerraClass Amazônia and MapBiomas**

Proceedings of the XIX Brazilian Remote Sensing Symposium · 2019

CAPANEMA, V. P.; SANCHES, I. D.; ESCADA, M. I. S. Comparison between land use and land cover thematic products from TerraClass Amazônia and MapBiomas: class adherence test. In: Brazilian Remote Sensing Symposium, 19th (SBSR), 2019, Santos. Proceedings... São José dos Campos: INPE, 2019. pp. 724-727. ISBN 978-85-17-00097-3.

<http://urlib.net/rep/8JMKD3MGP6W34M/3U67378>

### → **Mapping forest degradation intensity patterns**

Brazilian Journal of Cartography, v. 70, n. 1 · 2018

CAPANEMA, V. P.; PINHEIRO, T. F.; ESCADA, M. I. S.; SANT'ANNA, S. J. S. Mapping forest degradation intensity patterns: A case study in the Sinop region, Mato Grosso. Brazilian Journal of Cartography, v. 70, n. 1, pp. 199-225, Jan./Mar. 2018.

DOI: [10.14393/rbcv70n1-45254](https://doi.org/10.14393/rbcv70n1-45254)

### → **Forest Degradation in the Amazon: spatial indicator**

XVIII Brazilian Remote Sensing Symposium - SBSR · 2017

CAPANEMA, V. P.; ESCADA, M. I. S. Forest Degradation in the Amazon: Generation of a Spatial Indicator of Forest Degradation Susceptibility. In: XVIII Brazilian Remote Sensing Symposium - SBSR, Santos, 2017.

### → **Fire scars and land use change patterns in Pará**

XVIII Brazilian Remote Sensing Symposium - SBSR · 2017

SOUZA, G. M.; CAPANEMA, V. P.; ESCADA, M. I. S. Fire scars and land use and land cover change patterns in southwestern Pará, Brazil. In: XVIII Brazilian Remote Sensing Symposium - SBSR, Santos, 2017.

## VPC-GEOSER — Geospatial Solutions

**Co-Founder** of VPC-GEOSER, a startup specializing in transforming geospatial data into strategic decisions. The company develops **integrated geoprocessing, remote sensing, and data analytics solutions** applied to infrastructure, environmental management, and sustainability.

With a focus on **cloud technologies** and **scalable data architectures**, VPC-GEOSER delivers territorial monitoring systems, environmental impact analysis, predictive modeling, and interactive dashboards — connecting raw data to actionable insights for decision-makers.

**Vision:** Democratize access to geospatial intelligence, enabling public and private organizations to make data-driven decisions and contribute to territorial sustainability.

## Get in Touch

Have a project in mind? Send your message securely.

Your message...

Your email (optional)

 Send Message

Your data is sent privately.